

# Alternating Current (AC)

## # Faraday's Law :

Emf induced in any loop is equal to -ve of rate of change of magnetic flux

$$E_{ind.} = - \frac{d\phi_B}{dt} \Rightarrow Q_{ind.} = \frac{\Delta\phi_B}{R}$$

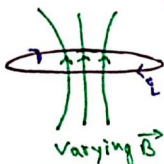
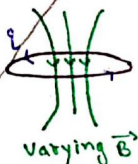
where,

$$\phi_B = \oint \vec{B} \cdot d\vec{s}$$

Magnetic flux  $\rightarrow$  No. of  $\vec{B}$  lines crossing an area join to the loop.

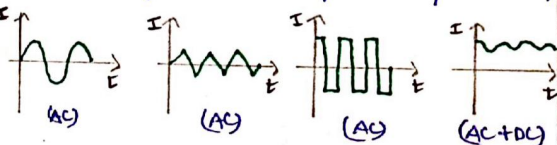
## # Lenz Law :

- $\rightarrow$  Informs about the dir<sup>n</sup> of emf / current
- $\rightarrow$  Based on conservation of energy
- $\rightarrow$  Effect will always oppose it cause



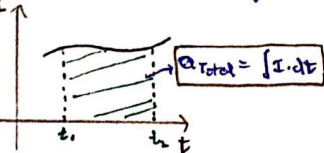
## # Cond<sup>n</sup> for a current/voltage to be alternating

- Amplitude is constant
- Half the cycle, Half -ve cycle
- Changes in both mag. & dir<sup>n</sup> periodically



## # Average/Mean Value :

$I_{avg.}$  is that const. DC current which send same charge in ckt in same time.



$$I_{avg} = \frac{Q_{total}}{\Delta t}$$

$$I_{avg} = \frac{\int_{t_1}^{t_2} I \cdot dt}{t_2 - t_1}$$

## # RMS/Effective value :

That DC which produce same amount of heat as that of AC in ckt in same time.

$$I_{rms} = \sqrt{\frac{\int_{t_1}^{t_2} I^2 \cdot dt}{t_2 - t_1}}$$

$$[\omega = \frac{2\pi}{T}]$$

$$[\omega = 2\pi f]$$

## # Important Results :

(\*) For a sinusoidal AC  $\Rightarrow$

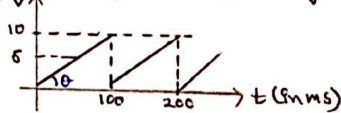
$$I_{avg} = \pm \frac{2I_0}{\pi} \quad [I = I_0 \sin \omega t]$$

(For half cycles)

$$\begin{aligned} (*) \quad I &= a \sin \theta & I_{max} &= a \\ I &= a + b \sin \theta & I_{max} &= a + b \quad (a, b > 0) \\ I &= a \sin \theta + b \cos \theta & I_{max} &= \sqrt{a^2 + b^2} \\ I &= a \sin \theta & I_{max} &= a \quad (a > 0) \end{aligned}$$

$$\begin{aligned} (*) \quad \langle \sin \theta \rangle &= \langle \sin 2\theta \rangle = 0 \\ \langle \cos \theta \rangle &= \langle \cos 2\theta \rangle = 0 \\ \langle \sin \theta \cdot \cos \theta \rangle &= 0 \\ \langle \sin^2 \theta \rangle &= \langle \cos^2 \theta \rangle = \frac{1}{2} \end{aligned}$$

Ex: Find avg. value of voltage.



Sol<sup>n</sup> Function is not given. So form one.

$$\begin{aligned} \therefore y &= mx + c \Rightarrow V = \tan \theta (t) + 0 \\ V &= \left( \frac{10}{100} \right) t \Rightarrow V = \frac{t}{10} \end{aligned}$$

Now,

$$V_{avg} = \frac{\int_0^{100} V \cdot dt}{100 - 0} \Rightarrow \frac{\int_0^{100} \frac{t}{10} \cdot dt}{10(100)} \Rightarrow \frac{(100)^2}{2 \times 10^3}$$

$$V_{avg} = 5V \quad \underline{\underline{Ans}}$$

Ex: Find effective current if DC = a & AC = b sin wt are superimposed.

$$\begin{aligned} \text{Sol<sup>n</sup> } I_{eff} &= I_{rms} = \sqrt{\langle (a + b \sin \omega t)^2 \rangle} \\ &= \sqrt{a^2 + b^2 \sin^2 \omega t + 2ab \sin \omega t} \\ &= \sqrt{a^2 + \frac{b^2}{2}} \quad \underline{\underline{Ans}} \end{aligned}$$

## # Phase & phase difference :

$$I = I_0 \sin (\omega t + \phi)$$

Initial phase =  $\phi$

Instantaneous phase =  $\omega t + \phi$

[Phase decides both value and sign]

Phase difference  $\Rightarrow$

$$V = V_0 \sin (\omega t + \phi_1)$$

$$I = I_0 \sin (\omega t + \phi_2)$$

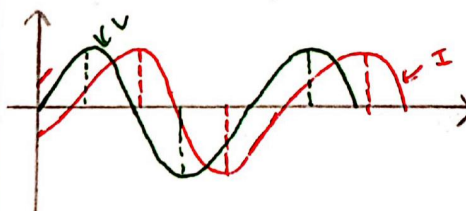
phase diff. of I w.r.t V =  $\phi_2 - \phi_1$

Phase diff of V w.r.t I =  $\phi_1 - \phi_2$

## # Lagging and Leading :

If V leads I  $\Rightarrow$  I lags V, then  $\rightarrow$  Then V reaches it max. before I

$$\begin{aligned} \text{If } V &= V_0 \sin \omega t \Rightarrow I = I_0 \sin (\omega t - \phi) \\ V &= V_0 \sin (\omega t - \phi) \Rightarrow I = I_0 \sin \omega t \end{aligned}$$



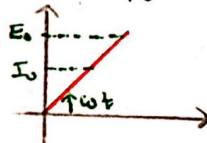
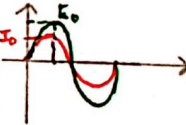
## # Purely Resistive Circuit :



$\rightarrow$  I & V in same phase  
 $\rightarrow$  No phase difference

$$\rightarrow I_0 = \frac{E_0}{R}$$

$$\rightarrow I_{rms} = \frac{E_{rms}}{R} = \frac{E_0}{\sqrt{2}R}$$



## # Purely Inductive Circuit :



$\rightarrow$  V leads I by  $\pi/2$

$$\rightarrow X_L = \omega L$$

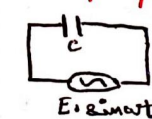
$$\rightarrow I_0 = \frac{E_0}{X_L} \quad I_{rms} = \frac{E_0}{\sqrt{2}X_L}$$



$$V = V_0 \sin \omega t$$

$$I = I_0 \sin (\omega t - \pi/2)$$

## # Purely Capacitive Circuit :



$\rightarrow$  I leads V by  $\pi/2$

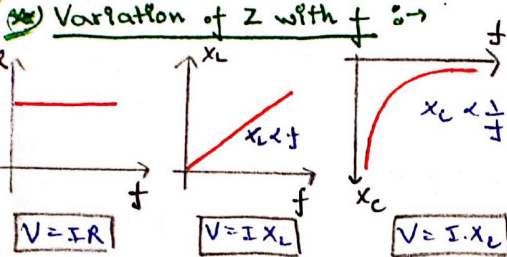
$$\rightarrow X_C = \frac{1}{\omega C}$$

$$\rightarrow I_0 = \frac{E_0}{X_C} \quad I_{rms} = \frac{E_0}{\sqrt{2}X_C}$$

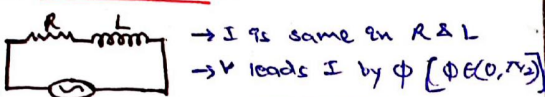


$$V = V_0 \sin \omega t$$

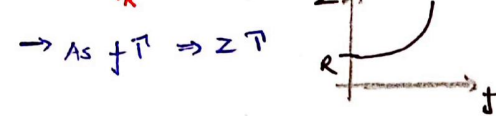
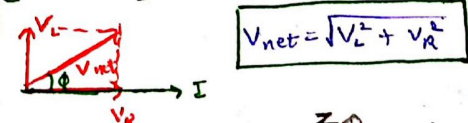
$$I = I_0 \sin (\omega t + \pi/2)$$



### # R-L Circuit :



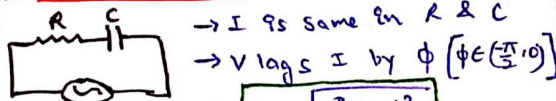
$\tan \phi = \frac{V_L}{V_R}$   $\rightarrow Z = \sqrt{R^2 + X_L^2}$



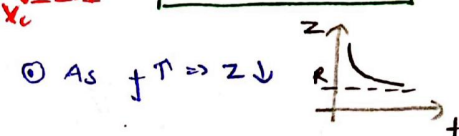
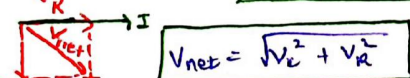
At very low  $f$  :  $Z = R$  ( $X_L \rightarrow 0$ )

At very high  $f$  :  $Z = X_L$

### # R-C Circuit :



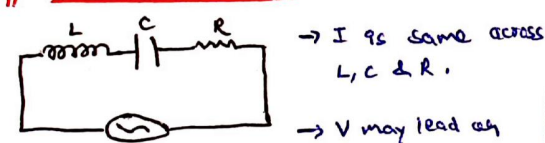
$Z = \sqrt{R^2 + X_C^2}$



At very low  $f$  :  $Z = X_C$

At very high  $f$  :  $Z = R$  ( $X_C \rightarrow 0$ )

### # L-C-R Circuit :



① When  $V_L > V_C$



$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$

### # Resonance :

- $\rightarrow$  When  $V_L$  &  $V_C$  become equal
- $\rightarrow$  When all voltage only across R
- $\rightarrow$  When no phase diff. b/w  $I$  &  $V$  in LCR ckt

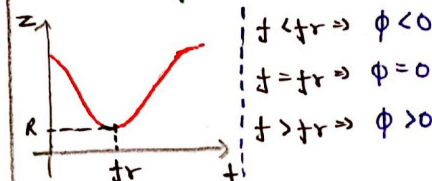
#### a) Cond<sup>n</sup> of Resonance $\Rightarrow$

- (i)  $X_L = X_C$  (iv)  $Z_{min} = R$
- (ii)  $V_L = V_C$  (v)  $I_{max} = \frac{V}{R}$
- (iii)  $\phi = 0$

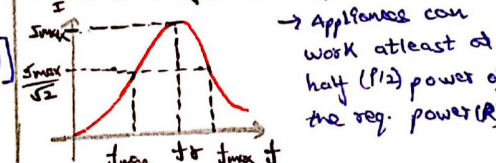
#### b) Resonance frequency $\Rightarrow$

$\omega_r = \frac{1}{\sqrt{LC}}$   $f_r = \frac{1}{2\pi\sqrt{LC}}$

#### c) Variation of Z with f $\Rightarrow$



#### d) Variation of I with f $\Rightarrow$



$\rightarrow$  Half power can be withdrawn at  $\frac{I_{max}}{2}$  current ( $\because P = VI$ )

Bandwidth  $\Rightarrow$  Range of 'f' for which any LCR ckt can practically work

$\Delta f = \frac{R}{L} = f_{max} - f_{min}$

Quality factor  $\Rightarrow$  Measures sharpness of resonance

$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$  &  $\frac{V_L}{R} = \frac{\omega_r L}{R}$   
 $\frac{V_C}{R} = \frac{1}{\omega_r C R}$

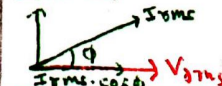
Selectivity  $\Rightarrow$  It's k<sup>y</sup> bandwidth kam, woh zyada selective, Kyunki kam f pe work krega.



Selectivity  $\propto \frac{1}{\text{Bandwidth (S)}}$

### # Power in AC Circuit :

$I_{rms} \cdot \sin \phi$   $\leftarrow$  Wattless current



Power =  $V_{rms} \cdot I_{rms} \cdot \cos \phi$   
 $Av_g = \frac{V_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cdot \cos \phi$

$\rightarrow$  'cos phi' is known as 'power factor'

$\cos \phi = \frac{P_{avg}}{P_{rms}} = \frac{R}{Z}$

$P_{avg} = P_{rms} \cdot \cos \phi$

### # Damped Oscillation :

Normal Oscillation  $\Rightarrow$

$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \Rightarrow \frac{d^2x}{dt^2} + \omega^2x = 0$

$x = A \sin \omega t$

Damped Oscillation  $\Rightarrow$

$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m}x = 0$

$x = A_0 e^{-b^2/2m} \cdot \sin(\omega t + \phi)$

$\rightarrow$  Amplitude decreases exponentially.

